

LAB 12

Firebase Database Setup & Implementation

Step 1: Firebase Setup (Configuration):

Packages

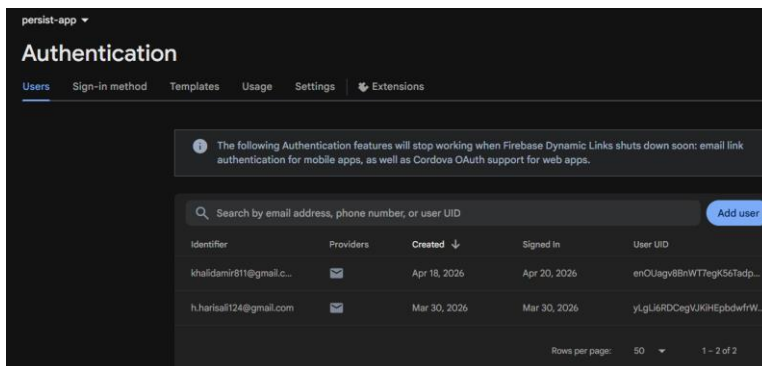
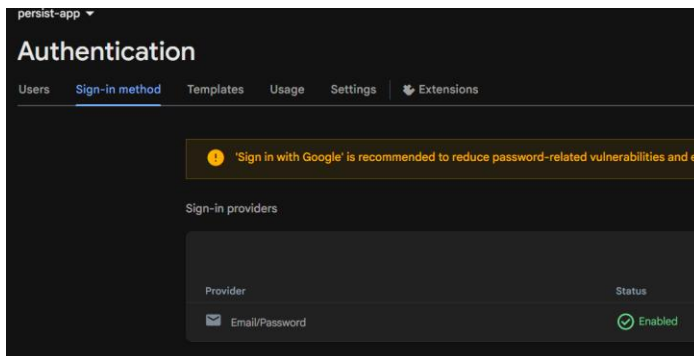
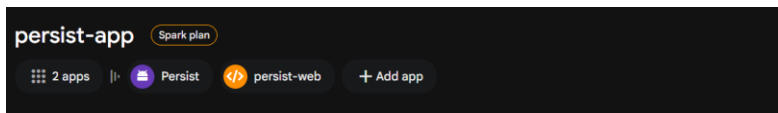
```
32   firebase_core: ^3.12.0
33   firebase_auth: ^5.5.0
34   cloud_firestore: ^5.6.0
```

Configuration

```
40   static const FirebaseOptions android = FirebaseOptions(
41     apiKey: 'AIzaSyAhlsghTEAX09XBUhJjiosMgGG6IxuDaZE',
42     appId: '1:880714616:android:69e9d9d4260f6026c1de8d',
43     messagingSenderId: '880714616',
44     projectId: 'persist-app-312ac',
45     storageBucket: 'persist-app-312ac.firebaseio.com',
46   );
```

Initialization:

```
11 void main() async {
12   WidgetsFlutterBinding.ensureInitialized();
13   await Firebase.initializeApp(options: DefaultFirebaseOptions.
14     currentPlatform);
15   runApp(const PersistApp());
16 }
```



persist-app

Cloud Firestore

Database

(default)

Ask Gemini about the core concepts to use Firestore

Data

Rules

Indexes

Disaster Recovery

Usage

Query Insights

Extensions

New in Cloud Firestore

More in Google Cloud

Panel view

Query builder

users

enOUagv8BnW...

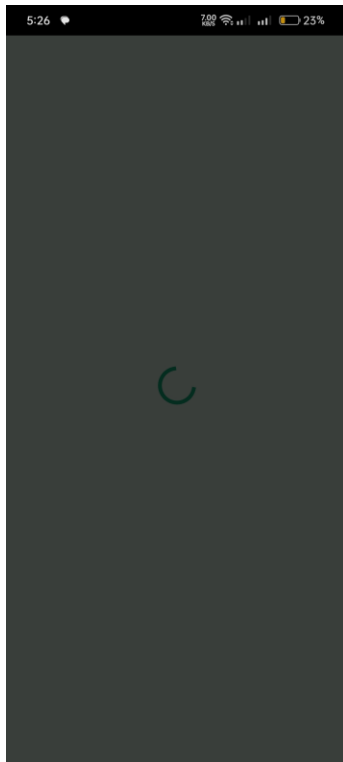
(default)	users	enOUagv8BnWT7egK56Tadpg2jXy2
+ Start collection	+ Add document	+ Start collection
users	enOUagv8BnWT7egK56Tadpg2jXy2	chats
	yLgLi6RDCegVJKiHEpbdwfrWrtD2	events
		+ Add field
		createdAt: April 18, 2026 at 3:35:39 PM UTC+5
		displayName: "haris"
		email: "khalidamir811@gmail.com"
		isPremium: false
		name: "haris"
		stabilityScore: 0
		streak: 0
		updatedAt: April 18, 2026 at 3:35:39 PM UTC+5

PHASE 2 — Firestore Data Models

User Profile

```
26 factory UserProfile.fromFirestore(DocumentSnapshot doc) {
27   final data = doc.data() as Map<String, dynamic>;
28   return UserProfile(
29     id: doc.id,
30     name: data['name'] ?? '',
31     email: data['email'] ?? '',
32     displayName: data['displayName'] ?? data['name'] ?? '',
33     createdAt: (data['createdAt'] as Timestamp?)?.toDate() ?? DateTime.now(),
34     updatedAt: (data['updatedAt'] as Timestamp?)?.toDate() ?? DateTime.now(),
35     isPremium: data['isPremium'] ?? false,
36     streak: data['streak'] ?? 0,
37     stabilityScore: data['stabilityScore'] ?? 0,
38   );
39 }
```

```
41 Map<String, dynamic> toFirestore() => {
42   'name': name,
43   'email': email,
44   'displayName': displayName,
45   'createdAt': Timestamp.fromDate(createdAt),
46   'updatedAt': Timestamp.fromDate(updatedAt),
47   'isPremium': isPremium,
48   'streak': streak,
49   'stabilityScore': stabilityScore,
50 };
51 }
```



Explain: As already logged in when opening the app the load sign sjows firbase read and let you logged in on the spot

Goal Model:

```
104     factory GoalModel.fromFirestore(DocumentSnapshot doc) {
105         final data = doc.data() as Map<String, dynamic>;
106         return GoalModel(
107             id: doc.id,
108             name: data['name'] ?? '',
109             category: data['category'] ?? 'Learning',
110             totalDays: data['totalDays'] ?? 0,
111             dueDate: data['dueDate'] ?? '',
112             completedDays: data['completedDays'] ?? 0,
113             streakDays: data['streakDays'] ?? 0,
114             active: data['active'] ?? true,
115             createdAt: (data['createdAt'] as Timestamp?)?.toDate() ?? DateTime.now(),
116             updatedAt: (data['updatedAt'] as Timestamp?)?.toDate() ?? DateTime.now(),
117             days: (data['days'] as List<dynamic>?) ?? [])
118                 .map((d) => DayModel.fromMap(d as Map<String, dynamic>))
119                 .toList(),
120         );
121     }
122
123     Map<String, dynamic> toFirestore() => {
124         'name': name,
125         'category': category,
126         'totalDays': totalDays,
127         'dueDate': dueDate,
128         'completedDays': completedDays,
129         'streakDays': streakDays,
130         'active': active,
131         'createdAt': Timestamp.fromDate(createdAt),
132         'updatedAt': Timestamp.fromDate(updatedAt),
133         'days': days.map((d) => d.toMap()).toList(),
134     };
135
```

2:40 PM

0.00 K/s 4G LTE 31

My Goals

Apr 19, 2026

+ New



1

Active



0%

Avg Progress



0

Best Streak

Active

All

Inactive

Learning



python

Due May 9, 2026

0% done

0/21 days

Today's tasks

- ☒ Work on: Goal progress
- ☐ Review progress
- ☐ Reflect & plan next step

View all 21 days



Home



Goals



Insights



Reflect



Settings